



BIOMEDICAL/ HEALTHCARE NER & RELATION EXTRACTION

(Thanks to Shweta Yadav)

Tools for Biomedical Entity Extraction

- MetaMap (2001), Apache cTakes (2010) : traditional architectures
 - Sentence boundary detector
 - Tokenizer
 - Normalizer
 - Part-of-speech (POS) tagger
 - Shallow parser
 - Named entity recognition (NER) annotator, including status and negation annotators.
- MedSpacy
- Various versions of LLMs

Aronson. *Effective mapping of biomedical text to the UMLS Metathesaurus: the MetaMap program*. AMIA Annual Symposium Proceedings, 2001.

H. Eyre, A. Chapman, K. Peterson, J. Shi, P. Alba, M. Jones, T. Box, S. DuVall, O. Patterson. *Launching into clinical space with medspaCy: a new clinical text processing toolkit in Python*. AMIA Annual Symposium Proceedings, 2022.

G. Savova, J. Masanz, P. Ogren, J. Zheng, S. Sohn, K. Kipper-Schuler, C. Chute. *Mayo clinical Text Analysis and Knowledge Extraction System (cTAKES): architecture, component evaluation and applications*. Journal of the American Medical Informatics Association, 2010.

[cTakes code](#)

Event Extraction from Unstructured Clinical Notes

with cTAKES^[1]

- extracts clinical concepts from text
- maps synonymous concepts to unique identifiers (CUI) in UMLS^[2] clinical knowledge graph

Patient with a history of diabetic foot and foot checkup. Patient states the toenails are thick, elongated, unable to trim on her own. Patient denies any recent history of trauma. Debrided nails today.

ANATOMY	DISEASE/DISORDER	SIGN/SYMPTOM	MEDICATION

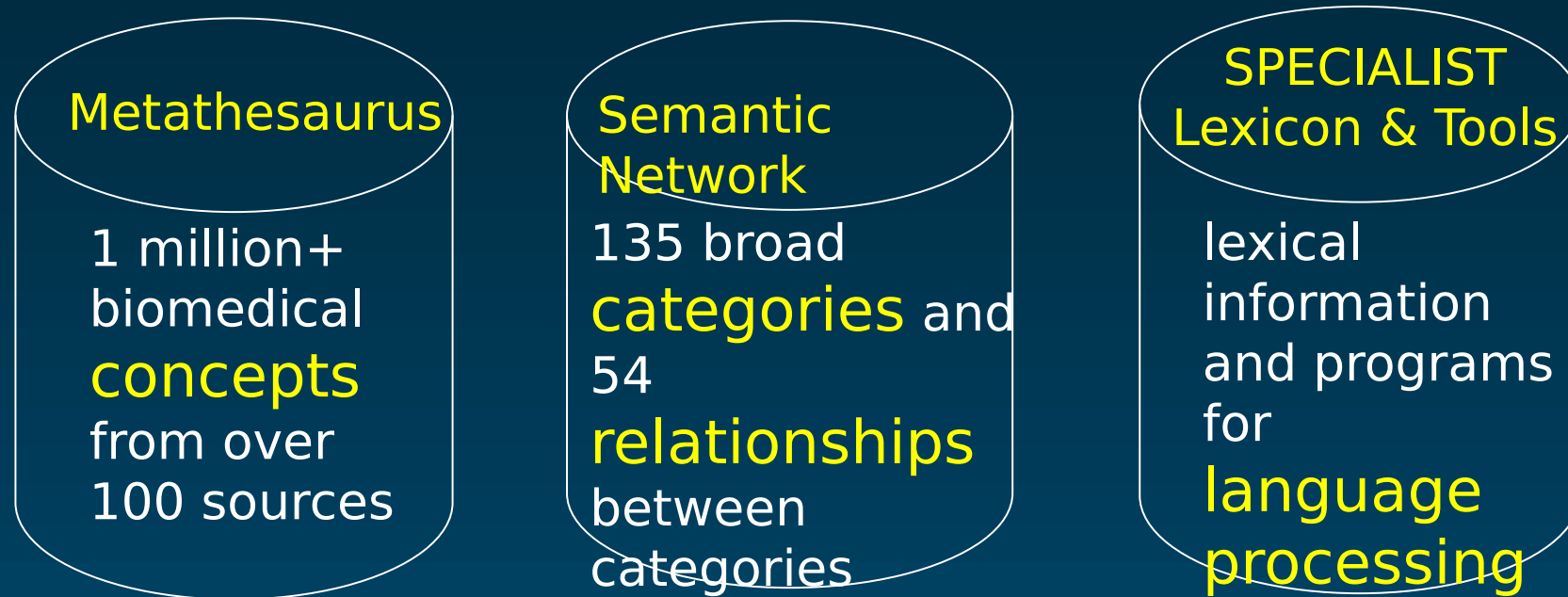
[1] Savova et al. "Mayo clinical text analysis and knowledge extraction system (ctakes): architecture, component evaluation and applications," JAMIA. 2010.
[2] Bodenreider "Using UMLS semantics for classification purposes." AMIA Symposium. 2000.

UMLS: Unified Medical Language System

2026AA Release

- 18M Names
- 3.5M Concepts
- 9.3M Codes
- 31 Languages (including Italian)
- 195 Vocabularies

UMLS consists of



3 Knowledge Sources

UMLS Objectives

- Intellectual middleware
- A set of multi-purpose tools for system developers
- Knowledge Sources used to overcome:
 - disparities in language and language format
Ex: atrial fibrillation, auricular fibrillation, af
 - disparities in granularity and perspective
Ex: Contusions, hematoma, bruise
Ex: Instruct patient to promptly report nosebleeds and excessive bruising (NIC), Epistaxis (MeSH)

Metathesaurus: clusters terms by meaning

- Concepts contain synonymous terms
- Preferred term is chosen (default can be changed)
- Unique identifier (CUI) is assigned

term	source	term type	source
ID			
Addison's disease	Metathesaurus	PN	
Addison's disease	SNOMED CT	PT	363732003
Addison's Disease	MedlinePlus	PT	T1233
Addison Disease	MeSH	PT	D000224
Bronzed disease	SNOMED Intl	SY	DB-70620
Primary Adrenal Insufficiency	MeSH	EN	D000224
Primary hypoadrenalism syndrome, Addison	MedDRA	LT	10036696

C0001403

Addison's disease

Organization of Concepts

Concept
C0001621

Term
L0001621

S0011231 Adrenal Gland Disease
A0020266 MeSH
A7568579 NCI Thesaurus
S0000441 Disease of adrenal gland
A0001264 SNOMED 1982
A6917004 SNOMED Clinical Terms
S0481705 Diseases of Adrenal Gland
A0014499 SNOMED 1982
S0220090 Diseases, adrenal gland
A0049924 MeSH

Term
L0181041

S0632950 Disorder of adrenal gland
A0688820 Read Codes
A4778687 SNOMED Clinical Terms
S0354509 Adrenal Gland Disorders
A6996540 MedlinePlus
A7576253 NCI Thesaurus
A7561794 Psychological Index Terms

Term
L1279026

S1520972 Nebennierenkrankheiten
A7500884

Semantic Network

■ 135 Semantic Types

- Broad subject categories (Clinical Drug, Virus)
- Ex:
 - **Addison's Disease**
 - **Semantic Type: Disease or Syndrome**

■ 54 Semantic Relationships

- Links between categories (isa, causes, treats)
- Ex:
 - Virus **causes** Disease or Syndrome

■ Types + Relationships

- Form the structure of the semantic network
- Broadly categorize the biomedical domain

54 Semantic Relationships

■ Hierarchical (isa = is a kind of)

■ among types

■ Animal *isa* Organism

■ Enzyme *isa* Biologically Active Substance

■ among relationships

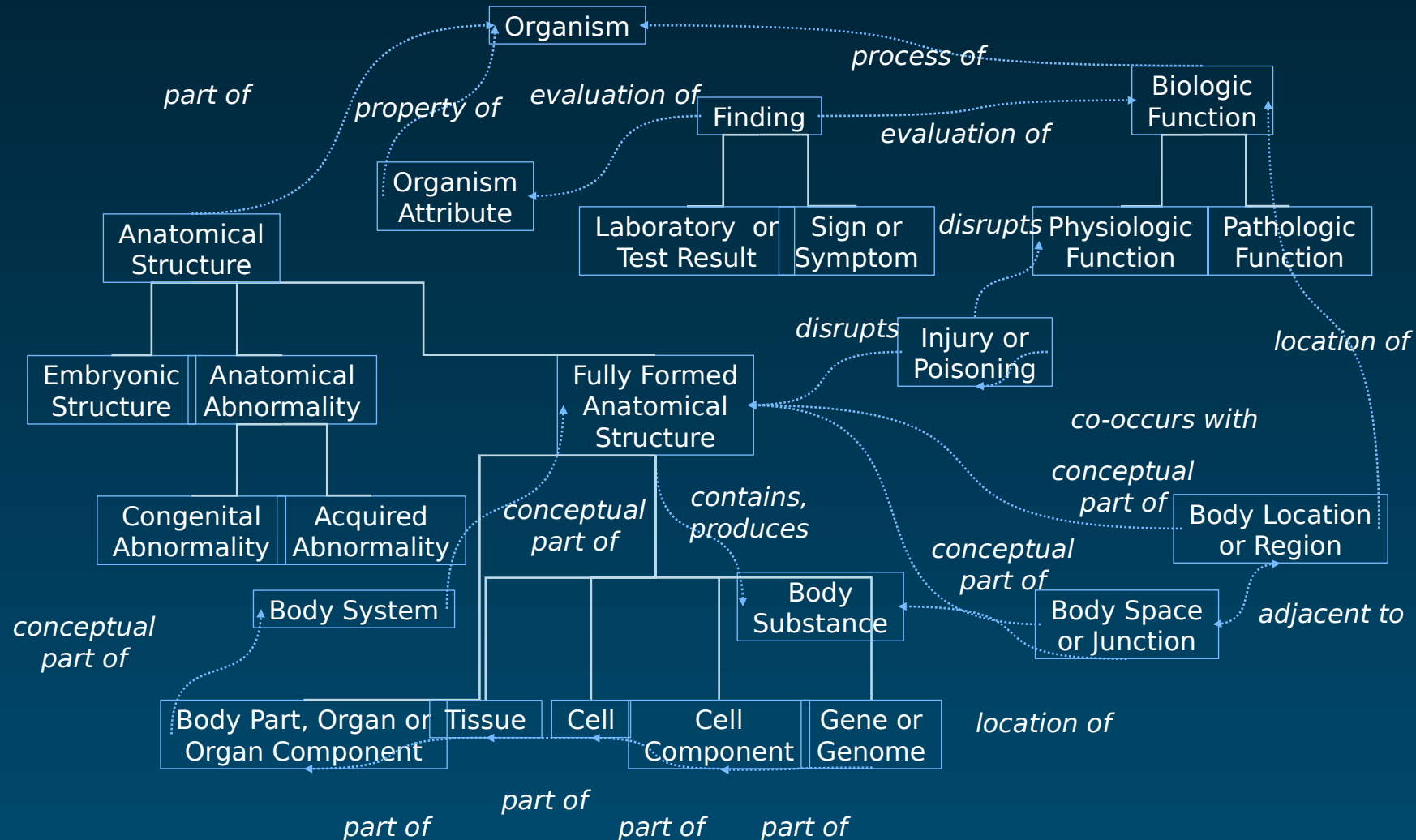
■ prevents *isa* affects

■ Non-hierarchical

■ Sign or Symptom *diagnoses* Pathologic Function

■ Pharmacologic Substance *treats* Pathologic Function

Semantic Network Relationships



Why have a Semantic Network?

■ Semantic Types

- **High level categories** assigned to Metathesaurus concepts
- Independent of position in source hierarchies

■ Semantic Relations

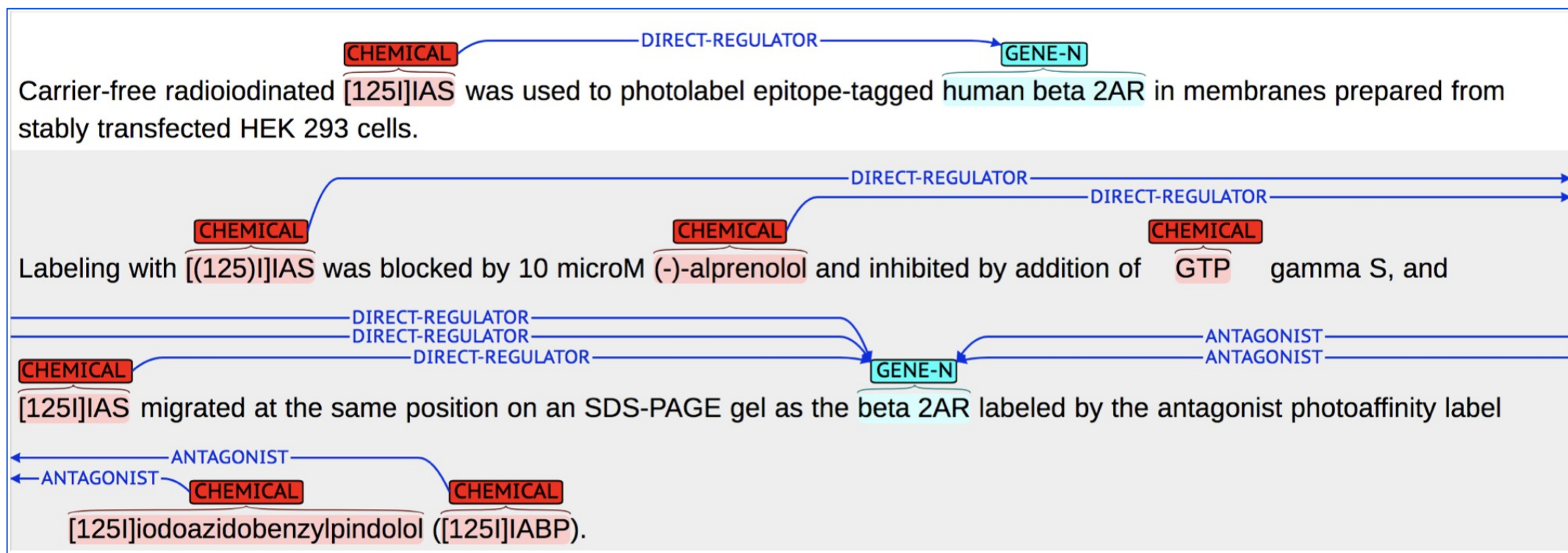
- **Useful links between** Semantic Types
- Relationships may hold at the concept level
- Other relationships may apply at the concept level

Common Biomedical Relation Extraction Tasks

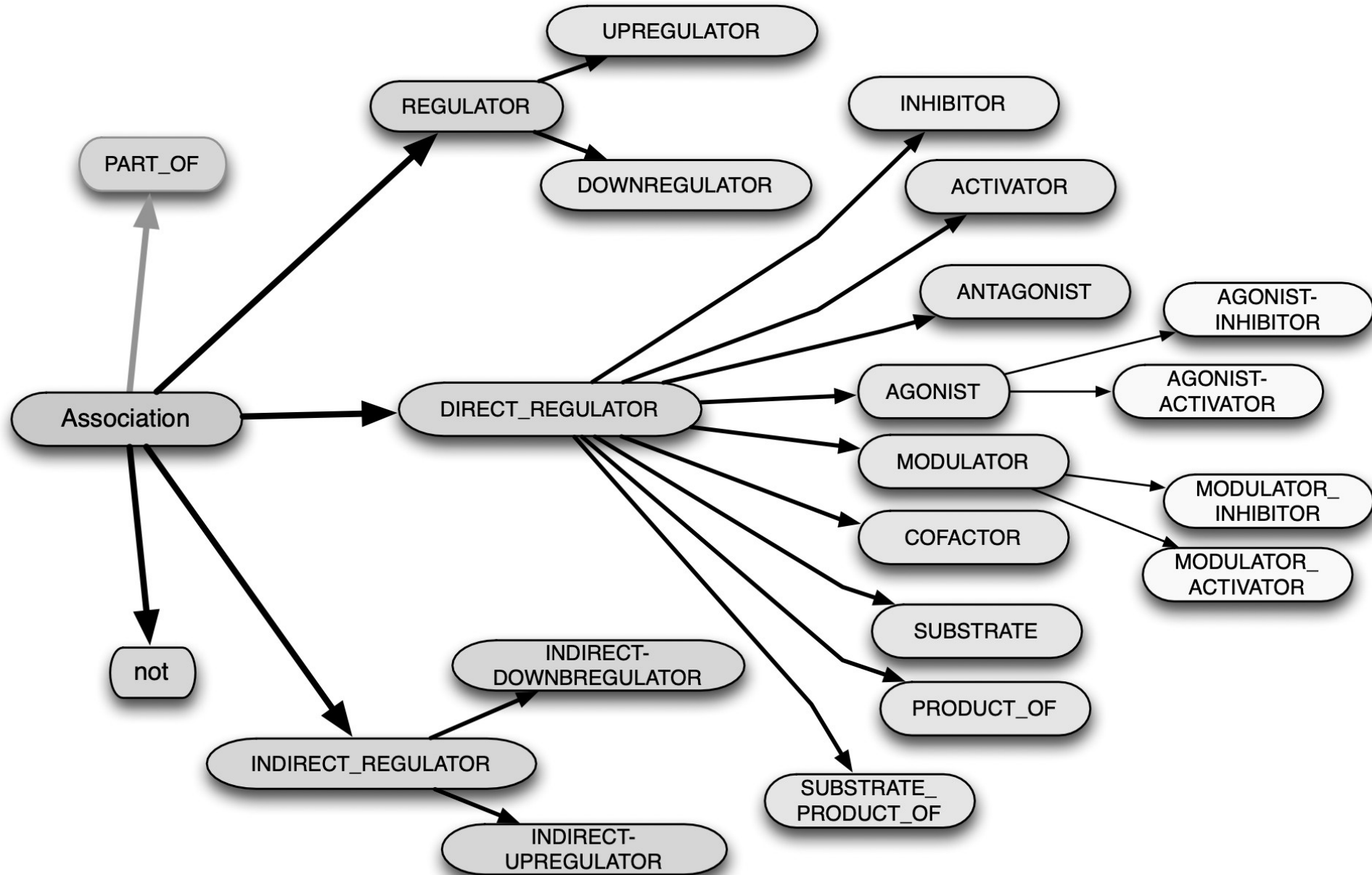
- Protein-protein interaction
- Chemical-protein interaction
- Gene-disease relations
- Chemical-disease relations

Chemical-protein interactions (CHEMPROT)

Aim: To automatically detect **relations** between **chemical compounds/drug** and **genes/proteins**.



ChemProt relation types

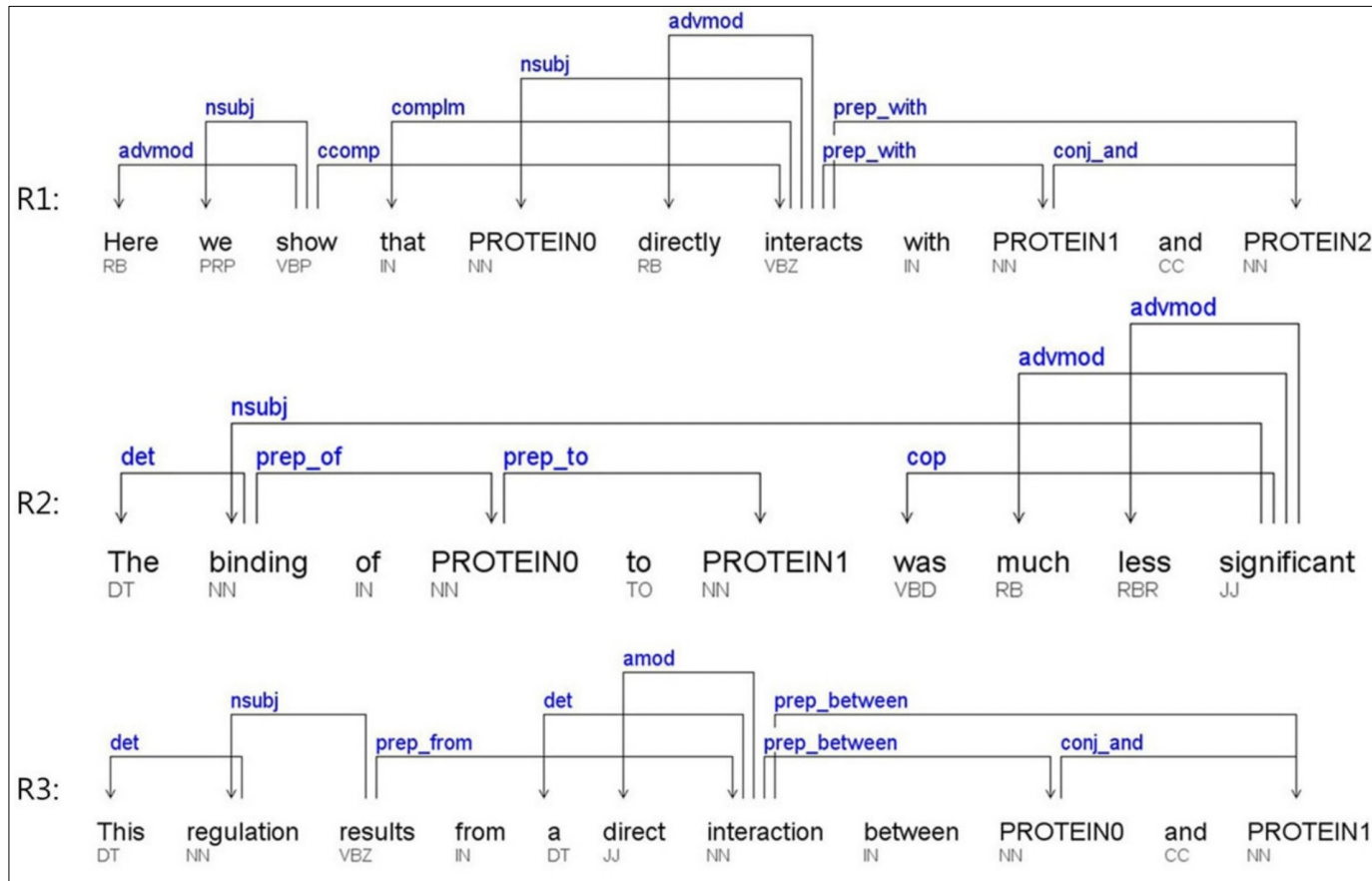


Medical Relation Extraction

- Medical problem - treatment relations:
 - Treatment improves medical problem
 - *hypertension* was controlled on *hydrochlorothiazide*.
 - Treatment worsens medical problem
 - *tumor* was growing despite the available *chemotherapeutic regimen*
 - Treatment causes medical problem
 - *Bactrim* could be a cause of these *abnormalities*
 - Treatment is administered for medical problem
 - *Lasix* periodically to prevent him from going into *congestive heart failure*
 - Treatment is not administered because of medical problem
 - *Relafen* which is contraindicated because of *ulcers*
- Medical problem - test relations
- Medical problem - medical problem relations

Existing Work on Biomedical Relation Extraction

- Rule-based



Rule 1: $P_i - REL - P_j$
Rule 2: $REL - of - P_i - PREP - P_j$
Rule 3: $REL - \{between, of\} - P_i - and - P_j$

Not realistic!
Long-distance relations in complex sentences are typically prone to parsing errors.

Existing Work on Biomedical Relation Extraction- Supervised Learning

Iron deficiency is the most common MND worldwide and leads to microcytic anemia , decreased capacity for work , as well as impaired immune and endocrine function .



Iron deficiency anaemia (IDA) and beta-thalassaemia are the most common causes of microcytic anaemia .



Studies here reported indicated that the anemia is hypochromic and microcytic anemia of blood loss and iron deficiency , in spite of the presence of large amounts of iron in the pulmonary tissue .



The high proportion of microcytic anaemia and the fact that gender differences were only seen after the menarche period in women suggest that iron deficiency was the main cause of anaemia .



MCV/RBC and (MCV)2 X MCH separated successfully the subjects with microcytic anaemia (heterozygous thalassaemia and iron deficiency) from normal controls .



Significantly higher serum homocysteine levels were reported in the iron deficiency anemia group compared to normal controls and in subjects with microcytic anemia and normal ferritin.



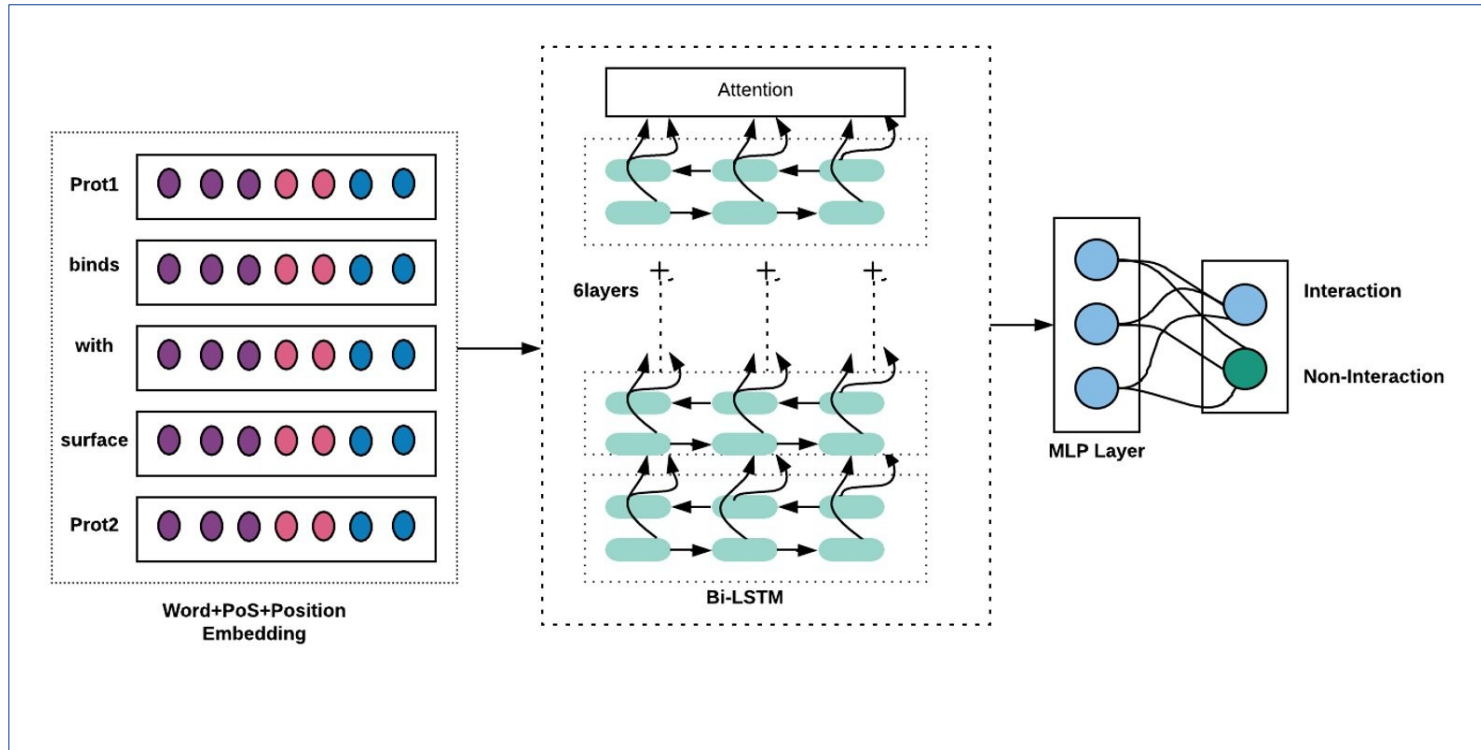
Idea: label examples, train a classifier (SVM, Naïve bayes, CRF)

Success! Better generalizability
Problem: require feature engineering:-()

Features

- **Semantic features**
 - path between the two entities in the dependency parse
- **Syntactic features**
 - the entities themselves
 - the types of the two entities
 - word sequence between the entities
 - number of words between the entities
 - path in the parse tree containing the two entities

Existing Work on Biomedical Relation Extraction



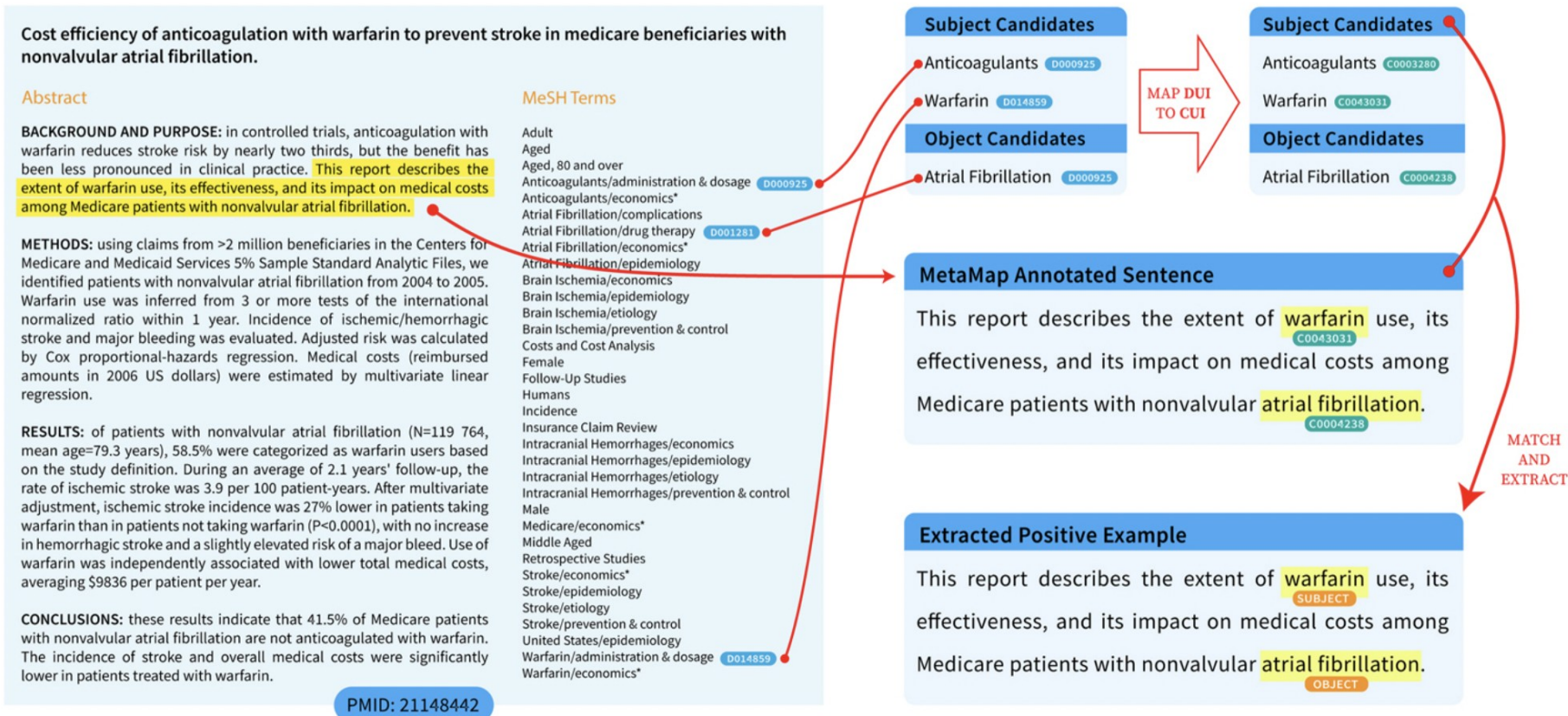
Idea: label examples, train a deep learning model (LSTM, RNN, GRU)

Success! Much better generalizability

Problem: require labelled data:-()

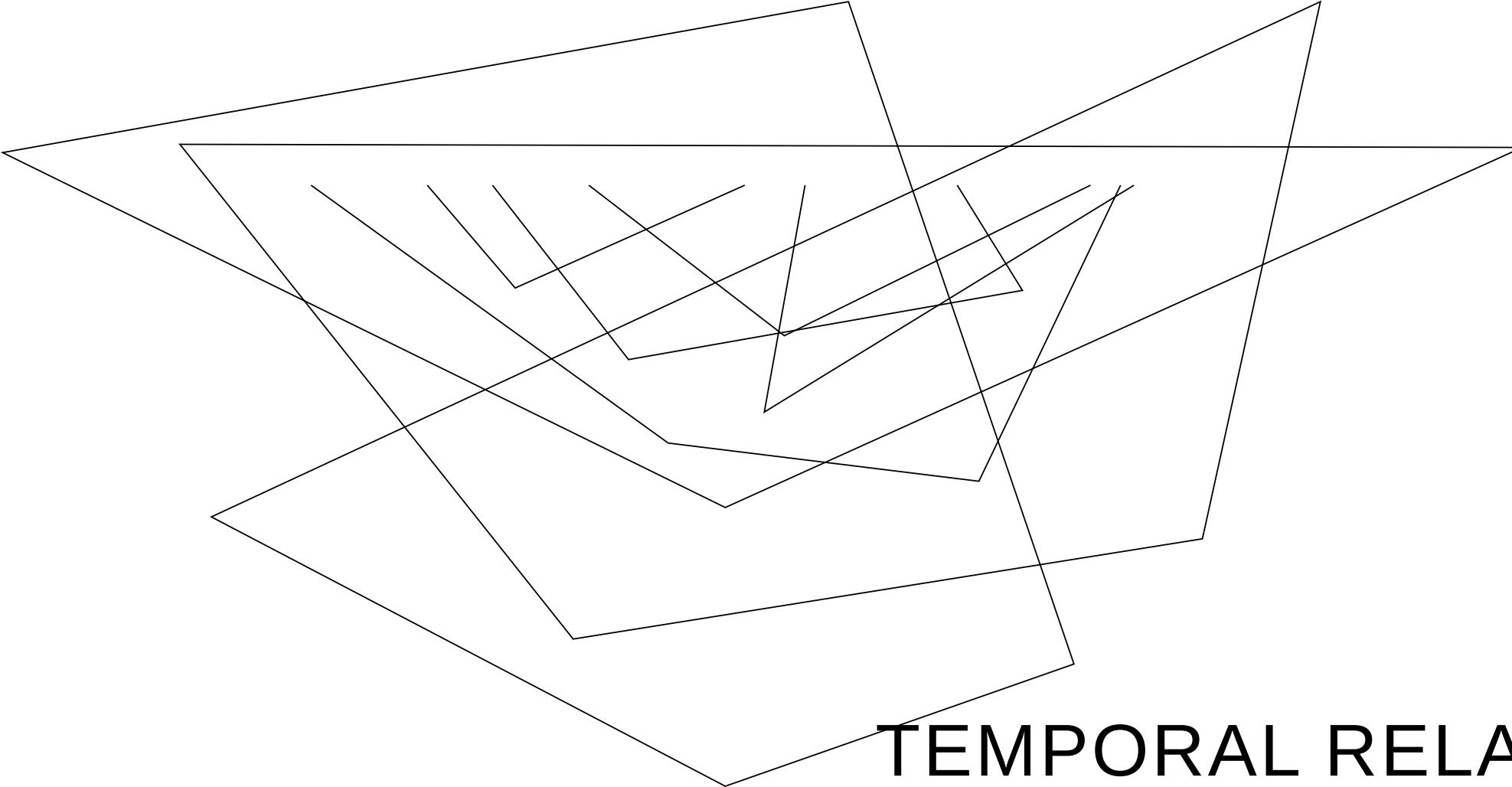
Existing Work on Biomedical Relation Extraction- Distant Supervision

- Idea: derive labels from an existing knowledge base (KB)
- Assume sentences with related entities are positive examples
- Assume sentences with unrelated entities are negative examples



Distant supervision: limitations

- Distant supervision is a powerful idea — but it has two limitations:
 1. Not all sentences with related entities are truly positive examples
 2. Need an existing KB to start from — can't start from scratch



TEMPORAL RELATIONS in CLINICAL TEXT

(again, thanks to Shweta Yadav & Rochana Chaturvedi)

Knee pain has increased over the last two or three weeks prior to the admission

Knee pain has increased over the last two or three weeks prior to the admission

EVENT **EVENT**

EVENT

- **PROBLEM** (e.g., disease, injury)
- **TREATMENT** (e.g., medication, therapeutic procedure)
- **TEST** (e.g., diagnostic procedure, laboratory test)
- **CLINICAL_DEPT** (i.e., the event of being transferred to a department)
- **EVIDENTIAL** (e.g., report, show)
- **OCCURRENCE** (i.e., all other events)

Knee pain has increased over the last two or three weeks prior to the admission

EVENT

TIMEX

EVENT

- DATE
- TIME
- DURATION
- FREQUENCY

Knee pain has increased over the last two or three weeks prior to the admission

EVENT

TIMEX

EVENT

- BEFORE
- AFTER
- SIMULTANEOUS
- OVERLAP
- BEGUN_BY
- ENDED_BY
- DURING
- BEFORE_OVER
LAP

Relazioni temporali

Graphical Representation	Allen's representation	TimeML schema	I2b2 corpus schema	THYME-TimeML schema (TLINK)	THYME-TimeML schema (DocTimeRel)
	X BEFORE Y	X BEFORE Y	X BEFORE Y	X BEFORE Y	X BEFORE Y
	X MEETS Y	X I_BEFORE Y	X BEF_OVERLAP Y	-	X BEF_OVERLAP Y
	X OVERLAPS Y	-	X OVERLAPS Y	X OVERLAPS Y	X OVERLAPS Y
	X DURING Y	X IS_INCLUDED Y	X DURING Y	-	-
	X STARTS Y	X BEGINS Y	-	X BEGINS_ON Y	-
	X FINISHES Y	X ENDS Y	-	-	-
	X FINISHED_BY Y	X ENDS_BY Y	X ENDS_BY Y	X ENDS_ON Y	-
	X AFTER Y	X AFTER Y	X AFTER Y	-	X AFTER Y
	X MEET_BY Y	X I_AFTER Y	-	-	-
	X OVERLAPPED_BY Y	-	-	-	-
	X CONTAINS Y	X INCLUDES Y	-	X CONTAINS Y	-
	X STARTED_BY Y	X BEGINS_BY Y	X BEGINS_BY Y	-	-
	X EQUALS Y	X IDENTIFIES Y X SIMULT. Y X HOLDS Y X HELD_BY Y	X SIMULT. Y	-	-

Credit: Y. Bonescki Gumiel, L. Silva e Oliveira, V. Claveau, N. Grabar, E. Cabrera Paraiso, C. Moro, D. Ribeiro Carvalho. 2021. *Temporal Relation Extraction in Clinical Texts: A Systematic Review*. ACM Comput. Surv. 54, 7 <https://doi.org/10.1145/3462475>

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Hypertension for 10 years.

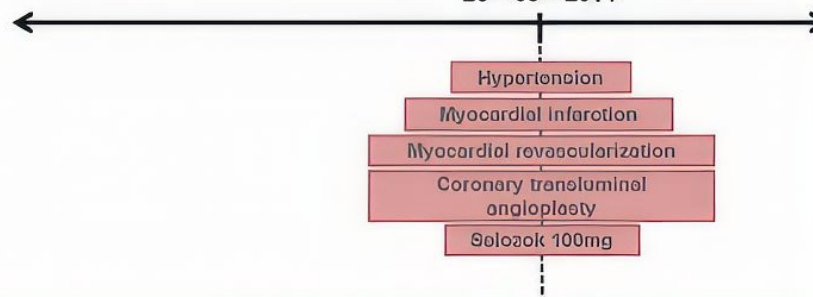
Myocardial infarction with myocardial revascularization in 2009.

Coronary transluminal angioplasty in 2013.

Plan: Selozok 100mg 1 tablet per day.

Simple

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DocTimeRel

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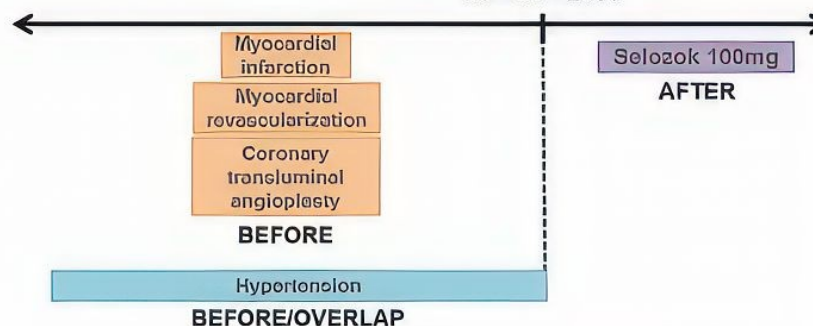
Hypertension
DocTimeRel: BEFORE/OVERLAP

Myocardial infarction with myocardial revascularization in 2009.
DocTimeRel: BEFORE

Coronary transluminal angioplasty in 2013.
DocTimeRel: BEFORE

Plan: Selozok 100mg 1 tablet per day.
DocTimeRel: AFTER

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DocTimeRel + TLINK

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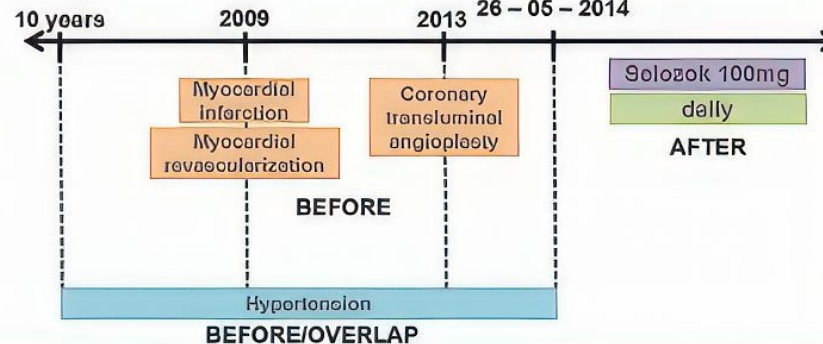
Hypertension
DocTimeRel: BEFORE/OVERLAP

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DocTimeRel: BEFORE

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DocTimeRel: AFTER

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RICONOSCERE RELAZIONI TEMPORALI IN FREE TEXT CLINICAL NOTES

